

Matthew Ngoy

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EDUCATION

Kean University

Union, NJ

Bachelor of Science in Computer Science (Cybersecurity Option); GPA: 3.52

Expected 2027

- **Relevant Courses:** Ethical Hacking; Digital Forensics; Computer Security; Cryptography & Cryptanalysis; Data Structures & Algorithms; Operating Systems; Database Management Systems; Applied Statistics; Linear Algebra; Software Engineering (in progress).
- **Honors & Programs:** NSF IUSE Scholar (2025, 2026); Dean's List; CAHSI Travel Scholar & Award Finalist; Upsilon Pi Epsilon; WiCyS (Secretary); Young Envoys Scholarship (YES) Fellow: 1 of 4 selected to address international delegates at CEAIE in Beijing.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, PHP, PowerShell

Security: OWASP Top 10, MITRE ATLAS, OSINT, Nmap, Wireshark, Metasploit, OpenVAS, Hashcat, Autopsy

Tools & Frameworks: Linux (Ubuntu, Kali), MySQL, SQLite, Node.js, PyTorch, scikit-learn, Arduino/ESP8266

EXPERIENCE

Cybersecurity Research Intern

May 2026 – Present

Kean University Center for Cybersecurity

Union, NJ / Wenzhou, China / Beijing, China

- Investigated and logged nonprofit cyber incidents via OSINT for the Non-Profit Cyber Incident Repository (NPCIR), a 400+ incident clearinghouse, standardizing each by CIA-triad impact, DHS CISA sector, and TAG Cyber taxonomy so researchers and nonprofit defenders can track sector-wide attack trends.
- Provisioned and secured a 30-laptop fleet for the Port Authority of NY/NJ STEaM Camp, running endpoint acceptance testing, wireless authentication configuration, and PowerShell SHA-256 file-integrity checks so students had identical, verified machines for every lab, workshop, and assignment.
- Built and debugged a browser-controlled IoT smart home lab (Arduino MEGA, ESP8266) for hands-on camp sessions, resolving floating-pin sensor readings, serial buffer overflows, and wireless configuration errors.
- Taught 12 lessons spanning cryptography, programming, and IP addressing to 60 high school students over the 3-week camp, providing live Wi-Fi and VM-access troubleshooting during sessions.

Teaching Assistant

Jan 2025 – Present

Kean University Department of Computer Science

Union, NJ

- Tutor Data Structures, DBMS, SQL, and Cybersecurity in 30-minute one-on-one drop-in sessions, 5 committed hours weekly, helping students turn failing code and stalled queries into working solutions; selected twice by CS faculty.

International Youth Delegate

Sep 2025 – Present

International Forum We, the Youth (IFWY), funded delegate fellowship

New York, NY / Seoul, Korea

- Spearheaded project design and scope for SelectSolar, an initiative to divert used solar panels from landfills and supply low-cost, energy-efficient power to schools and non-profits.
- Represented North America as 1 of 25 delegates at Cornell Tech and the IFWY Final Conference in Seoul, collaborating on international environmental policy and sustainable solutions.

PROJECTS

Red-Team Harness for Proof-of-Inference Consensus | Python, SQLite, OpenAI API

June 2026 – Present

- Red-teamed HadAgent, a blockchain that verifies AI model outputs by having independent nodes recompute and vote on them; uncovered that the vote proves only that an answer is reproducible, not that it is correct or safe.
- Engineered a simulator of the published design (Ed25519-signed records, Merkle blocks, trust tiers) with 5 attack runners; reproduced a validation bug from the project's history that admitted every malformed record (30 of 30).
- Evaluated 28 OWASP LLM Top 10 / MITRE ATLAS-tagged attacks against Llama 3.2 3B using an LLM-as-judge ported from "Debate Coach"; the in-path safety defense cut attack success from 39% to 14%.

RESEARCH & PRESENTATIONS

NSF IUSE Research Scholar, "Where Is Gen Z Living?" | MySQL, PHP, JS

May 2026 – Present

- Led a study correlating young-adult demographics with housing affordability; built a SQL data mart of 3,114 county-level records and a 3-tier MySQL/PHP/JS dashboard computing Pearson correlations and regression trendlines for 6 variable pairs; accepted to the 2026 GMIS Research Poster Competition (Albuquerque, NM).

"Debate Coach" LLM Evaluation, CRA UR2PhD Program | JS, Node.js, OpenAI API

Feb – Apr 2026

- Co-designed a 10-week evaluation of LLM debate coaches (GPT-4-as-judge, 12-participant pilot); fine-tuning tripled citation density but fabricated 100% of 336 citations, motivating RAG; presented at the UR2PhD Showcase.

NSF IUSE Research Scholar, "Fine-Tuned LLM Framework" | First Author, GMIS 2025

May – Oct 2025

- Built the fine-tuning module for an AI tutoring platform (Node.js preprocessing, token-budget validators); in side-by-side evaluations across 9 learner profiles, fine-tuned GPT-3.5-Turbo replaced the base model's generic answers with learner-matched Socratic scaffolding; sole presenter at GMIS and Kean Research Days.